



Using the Legacy BPL Editor

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1

Introduction to the Legacy BPL Editor

This page introduces the legacy BPL Editor (the **Business Process Designer** page), which enables you to create BPL [business processes](#) for interoperability [productions](#).

Note: Starting with 2026.1, the product includes a new BPL Editor. To access this application, click *Open in the new UI* on the existing BPL Editor. See [Introduction to the BPL Editor](#).

1.1 Accessing the Editor

To access this page:

1. Log in to the Management Portal.
2. Click **Interoperability > Build > Business Processes**.

Important: After a period of inactivity, the Management Portal may log you out and discard any unsaved changes. Inactivity is the time between calls to the InterSystems IRIS server. Not all actions constitute a call to the server. For example, clicking **Save** constitutes a call to the server, but typing in a text field does not. Consequently, if you are editing a business process, but have not clicked **Save** for longer than **Session Timeout** threshold, your session will expire and your unsaved changes will be discarded. After a logout, the login page appears or the current page is refreshed. For more information, see Automatic Logout Behavior in the Management Portal.

1.2 Areas of the Page

When you display the BPL Editor, it shows the last BPL you opened in this namespace, if any. This page has the following areas:

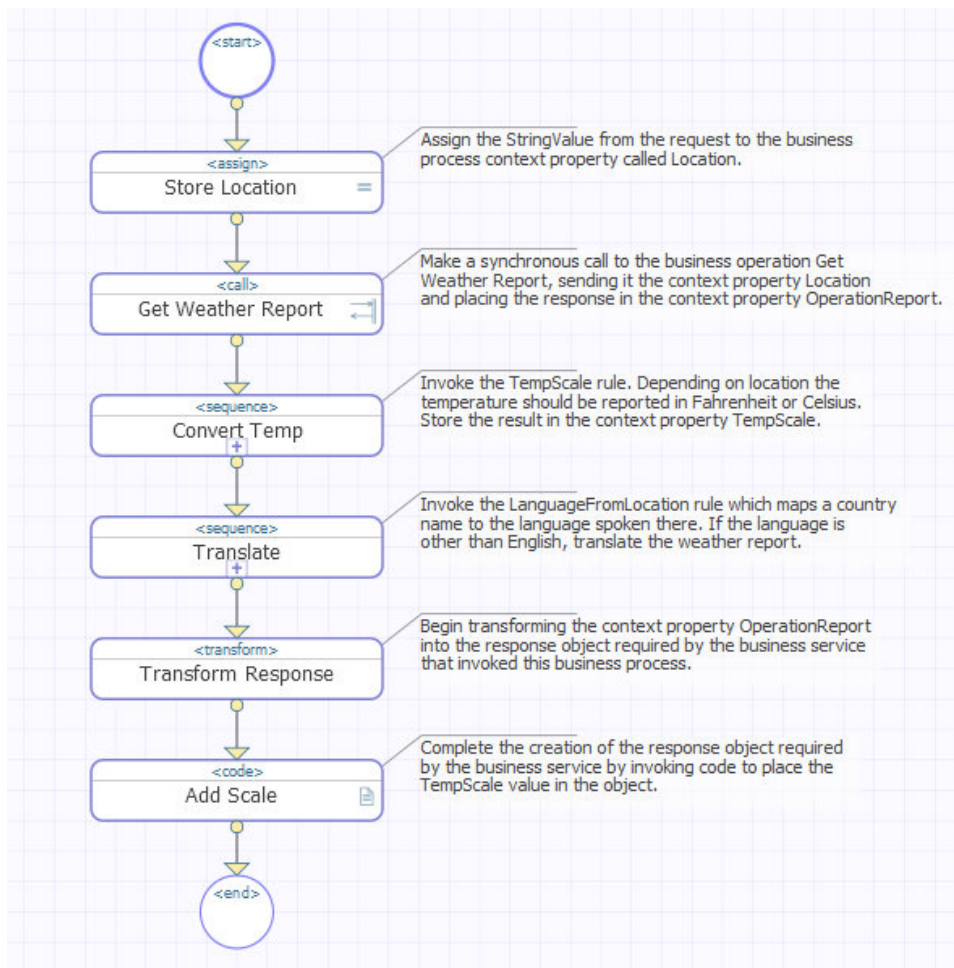
- The ribbon bar at the top displays options you can use to create and open BPL processes, compile the currently displayed BPL, change the zoom display of the diagram, add activities to the diagram, and so on.
- The left area displays the [BPL diagram](#).
- The right area displays the following tabs:

- **General**—contains settings for the overall definition of the BPL business process. See [Setting General Properties of the BPL Business Process](#).
- **Context**—enables you to [define the context object](#) for this BPL business process.
- **Activity**—contains settings for the selected item in the BPL diagram; see [Adding Activities to a BPL Diagram](#).
- **Preferences**—contains settings pertaining to the appearance of the BPL diagram. See [Setting BPL Diagram Preferences](#).

You can expand and collapse the right area using the double arrow icons.

1.3 The BPL Diagram

A BPL diagram is the editable, graphical representation of the logic defined by the business process. The following is an example.



A BPL business process consists of a connected set of activities, shown as different shapes in the diagram. Activities can use values received by the BPL and make decisions based on them, they call other business components, they can manipulate data, and they can call custom code. The process of creating a BPL business process consists primarily of defining the activities it includes and connecting those activities.

1.4 See Also

- [Creating BPL Business Processes](#)
- [Editing a BPL Diagram](#)

2

Creating BPL Business Processes (Legacy UI)

This page describes at a high level how to use the [legacy BPL Editor](#) to create and edit BPL [business processes](#) for interoperability [productions](#).

For information on performing these tasks with the new UI, see [Creating BPL Business Processes](#).

2.1 Creating a BPL Business Process

To create a BPL business process, do the following in the BPL Editor:

1. Click **New**.
This displays a dialog box.
2. Specify some or all of the following information:
 - **Package** (required)—Enter a package name or click the arrow to select a package in the current namespace.
Do not use a reserved package name; see [Reserved Package Names](#).
 - **Name** (required)—Enter a name for your BPL business process class.
 - **Description**—Enter a description for your BPL business process class; this becomes the class description.
3. Click **OK**.

The start and end points of the BPL diagram display in the [BPL Editor](#), ready for you to add activities to your BPL business process.

2.2 Opening a BPL Business Process

To open a BPL business process, do the following in the BPL Editor:

1. Click **Open**.

If you are currently viewing a BPL and you have made changes but have not yet saved them, InterSystems IRIS prompts you to confirm that you want to proceed (which will discard those changes).

2. Click the package that contains the BPL.

Then click the subpackage as needed.

3. Click the BPL class.
4. Click **OK**.

2.3 Setting General Properties of the BPL Business Process

Each BPL business process has a small set of properties that apply to the process as a whole. To set these properties, do the following in the BPL Editor

1. Click the **General** tab.
2. Modify the following settings:
 - **Language**—Can be **Python** or **ObjectScript**.
 - **Layout**—Select either **Automatic** or **Manual** for the size of the diagram. If you select **Manual** you can enter a **Width** and **Height**.
 - **Annotation**—Enter text to include in the class description.
 - **Includes**—An optional comma-delimited list of include file names, so that you can use macros in your <code> segments.
 - **Version**—Enter an optional version number of the BPL diagram
 - **Is component**—If true, include this process in the component library where it can be called by other processes.

See [<process>](#) for details on these properties.

2.4 Defining the context Object

Each BPL business process has a context object that provides information available for use in the BPL logic. To define this object, do the following in the BPL Editor:

1. Click the **Context** tab.
2. Modify the following settings:
 - **Request Class**—Choose the class of the incoming request for this process.
 - **Response Class**—Choose the class of the response returned by this process.
 - **Context Superclass**—Use this option to provide custom context properties, in a different way than adding to the **Context properties** list, described next. To use **Context Superclass**, create a custom subclass of `Ens.BP.Context`. In this subclass, define class properties to use as context properties. Use the name of this class as the value of

Context Superclass in the business process. Then when you create <assign> actions, for example, you can choose these custom properties in addition to the standard properties of the *context* object.

3. To add a property, click the plus sign next to **Context properties**.

Then enter values in the following fields:

- **Property Name**—Must be a valid identifier.
- Choose if the property data is one of the following: **Single Value**, **List Collection**, or **Array Collection**
- **Property Type**—Type of this property including parameters.
Enter a data type class name in the **Type** field or click the magnifying glass to browse for a class you want to use as a data type.
- **Default Value** (ignored for collections)—Enter an initial expression for a single value data type.
- **Instantiate**—Select this check box for object-valued properties if you want the object to be instantiated when it is created.
- **Description**—Enter an optional description of the context property.

4. Click **OK** to save your changes or click **Cancel** to discard them.

To set property parameters such as `MINVAL`, `MAXVAL`, `MINLEN`, `MAXLEN`, or others, add data type parameters to a *context* property when you first add the property, or at any subsequent time, by inserting a comma-separated list of parameters enclosed in parentheses after the data type class name. That is, rather than simply entering `%String` or `%Integer`, you can enter data types such as:

```
%String(MAXLEN=256)
%Integer(MINVAL=0, MAXVAL=100)
%String(VALUELIST=", Buy, Sell, Hold")
```

Once you have defined properties of the *context* object, you can refer to them anywhere in BPL using ordinary dot syntax and the property name, as in: `context.MyData`

For reference details, see the following resources:

- Typically you choose the property type from types described in Data Types. These include `%String`, `%Integer`, `%Boolean`, and so on.
- System data types have optional parameters. For details, see Parameters. These include the `MINLEN` and `MAXLEN` parameters that set the minimum and maximum allowed lengths of a `%String` property. The default maximum `%String` length is 50 characters; you can reset this by setting the `MAXLEN` for that `%String` property to another value.

By default, the *ruleContext* passed to the rule is the business process execution context. If you specify a different object as a context, there are some restrictions on this object: It must have a property called `%Process` of type `Ens.BusinessProcess`; this is used to pass the business process calling context to the rules engine. You do not need to set the value of this property, but it must be present. Also, the object must match what is expected by the rule itself. No checking is done to ensure this; it is up to the developer to set this up correctly.

2.5 Adding an Activity

Each BPL business process consists of a set of connected activities. In general, to add an activity to a BPL business process, do the following in the BPL Editor:

1. Click **Add Activity** in the ribbon bar.

This adds a new shape to the BPL diagram.

2. Edit the activity details in the **Activity** tab on the right.
3. Use drag and drop options to modify how this activity is connected to others.

For details, see [Adding Activities to BPL](#) and [Editing a BPL Diagram](#).

2.6 Undoing a Change

To undo the previous change in the BPL Editor, click the Undo button .

2.7 Saving a BPL

To save a BPL while in the BPL Editor, do one of the following:

- Click **Save**.
- Click **Save As**. Then specify a new package, class name, and description and click **OK**.
- Click **Compile**. This option saves the BPL and then compiles it.

2.8 Compiling a BPL

To compile a BPL while in the BPL Editor, click **Compile**. This option saves the BPL and then compiles it.

2.9 See Also

- [Editing a BPL Diagram](#)

3

Editing a BPL Diagram (Legacy UI)

Each BPL business process consists of a set of connected activities, represented by shapes in the BPL diagram. This page describes how to make changes in a BPL diagram, within the [legacy BPL Editor](#).

For information on performing these tasks with the new UI, see [Editing a BPL Diagram](#).

3.1 Basics

- To select an activity, click it. When you do so, its attributes display in the **Activity** tab, where you can edit their values.
- To select multiple activities, hold down the **Ctrl** key while clicking.
- To clear the selection of a selected activity, click it again.
- You can [connect](#) activities via drag and drop actions.
- The toolbar provides options for adding activities; cutting, copying, pasting, deleting activities; and undoing changes.
- The BPL Editor automatically validate activities as you add them to the diagram. If it detects an element with a logical error, it displays a red warning on the **Activity** tab for the element along with the reason for the error.
- Several kinds of BPL activities are displayed as their own subdiagrams. To see them in detail, it is necessary to [drill down](#) into them.

3.2 Color Indicators

The BPL Editor provides the following color indicators for shapes in the diagram:

- Typically the interior color of a BPL diagram shape is white, with a blue outline. If you select the shape, its interior color changes to yellow and the outline becomes bolder.
- If the shape is in error, its outline is red.
- If the shape is disabled, its interior color is gray, with a gray outline. When you select a disabled shape it shows a dotted outline.

Also, when a shape represents a complex activity such as <if> or <switch> that has multiple branches, joins, or other types of related shapes elsewhere in the BPL diagram, clicking on one of these shapes highlights the related shapes in green with a purple outline.

3.3 Specifying Diagram Preferences

The **Preferences** tab contains the following settings that apply to the appearance of the BPL diagram:

- **Gridlines**—Select one of the following choices for the appearance of the grid lines on the diagram: **None**, **Light**, **Medium**, or **Dark**.
- **Show annotations**—Reveal or hide the text notes that explain each shape. When you reveal annotations, they appear to the upper right of each shape that has an <annotation> element in the BPL document.
- **Auto arrange**—Cause each new shapes in the diagram to automatically conform to a structured arrangement without needing to select  after adding each shape.

Changing the position of shapes does not change the underlying BPL code.

3.4 Adding an Activity

To add an activity to a BPL diagram, do the following:

1. Select an option from the **Add Activity** list.

This immediately places a new, unconnected activity to the diagram.

2. While the new activity is still selected, specify values as needed on the **Activity** tab:

- **Name**—Enter a name for the caption inside the shape.
- **Disabled**—Optionally select this check box to disable the activity; clear it to enable. The default is enabled.
- **Annotation**—Optionally enter text to appear as comments next to the shape in the diagram.

Other details depend on the type of activity. See the [BPL Reference](#).

3. [Connect](#) this activity to other activities as needed.

Or if this activity should be inserted between two existing activities, do the following:

1. Select the connector that connects those two activities.
2. Select an option from the **Add Activity** list.

This immediately inserts the new activity between the two other activities, connected to both of them.

3. While the new activity is still selected, specify values as needed on the **Activity** tab, as described above.

3.4.1 Adding a Call Activity

A common task in a BPL business process is to add a Call activity. The following information is necessary to properly create a new <call> to one of the available business processes or business operations in the production:

- Input
- Output
- Name
- Target

- Request

3.5 Editing Properties of an Activity

To edit properties of an activity, do the following:

1. Click the activity.
2. Specify values as needed on the **Activity** tab:
 - **Name**—Enter a name for the caption inside the shape.
 - **Disabled**—Optionally select this check box to disable the activity; clear it to enable. The default is enabled.
 - **Annotation**—Optionally enter text to appear as comments next to the shape in the diagram.

Other details depend on the type of activity. See the [BPL Reference](#).

3.6 Removing an Activity

To remove an activity from a BPL diagram, do the following:

1. Select the activity.
2. Click the Remove  button in the toolbar.

3.7 Adding a Connection

Each activity is displayed with one triangular *input* point and one *output* circle. You use these when connecting activities.

To add a connection from one activity to another, do the following:

1. Click the output circle of one activity.
2. Drag the cursor to input triangle of the other activity and then release.
Equivalently, you can click the input triangle of one activity and drag to the output circle of the other activity.
3. Optionally select the connector, and enter a name for it on the **Activity** tab.

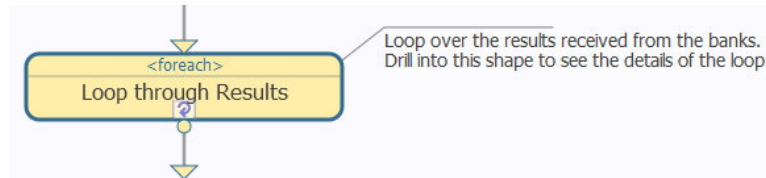
The tool does not allow you to make an illegal connection.

Once two shapes are connected, the connection is preserved no matter where you drag the respective shapes. You can drag shapes to any layout position you wish, within the same diagram.

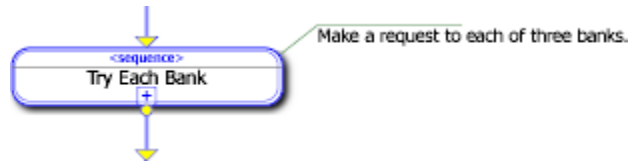
3.8 Drilling Down and Back

Several kinds of BPL activities are displayed as their own subdiagrams. To see them in detail, it is necessary to drill down into them. Examples include `<foreach>` and `<sequence>`. The main BPL diagram shows only a stub, with an indicator that you need to click for more details.

For example, the following shows a `<foreach>` loop, as you would see it within the main BPL diagram:



The cyclical arrow icon at the bottom of the shape is a reminder that there are details not shown here. Another example is `<sequence>`, which displays a plus sign icon to remind you that there are more details:



In all cases, to drill down, you can do either of the following:

- Click the icon (circular arrow or plus sign).
- Click the activity and then click the Drill Down icon  in the toolbar.

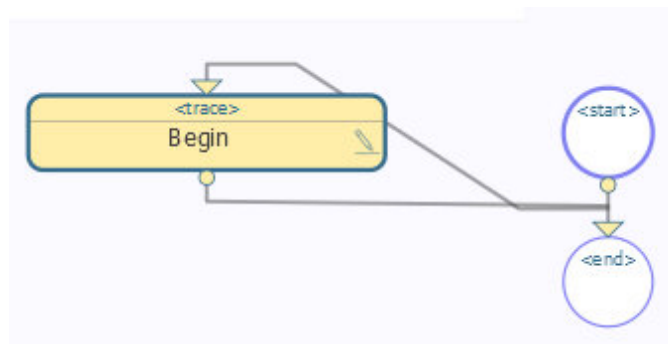
The BPL Editor then displays the full details for that activity, from start to end.

To return to the higher logical level, click the Drill Up icon  in the toolbar.

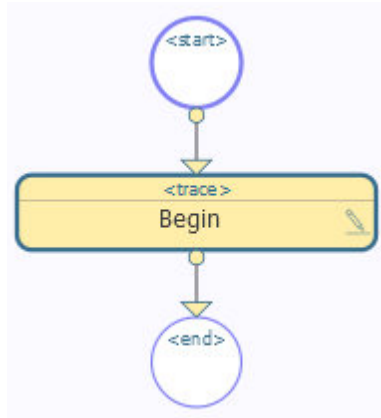
3.9 Adjusting the Layout

After you add shapes or create new connections, you can tidy the diagram by clicking the arrange icon  on the tool bar.

For example, if you do not have the auto arrange feature set in your preferences, when you add a shape to a BPL diagram it looks something like the following figure.



When you click the auto arrange tool  the shapes are aligned as shown in the following figure.



If you want your diagrams to always use this type of structured layout, select the **Auto arrange** check box on the **Preferences** tab.

By default, when you open a BPL diagram for the first time, the auto arrange feature is enabled. This choice may or may not be appropriate for a particular drawing. You can disable automatic arrangement to ensure that your diagram always displays with exactly the layout you want by clearing the **Auto arrange** check box on the **Preferences** tab. This way, when the diagram is displayed, it does not take on any layout characteristics except what you have specified.

3.10 See Also

- [Creating BPL Business Processes](#)

